

MODEL QUESTION PAPER 2

S4 INSTRUMENTATION ENGINEERING

SUBJECT: PROCESS VARIABLE MEASUREMENTS

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	List two units of pressure.	M1.01	U
2	Calculate the gauge pressure if the atmospheric pressure is 760 mm of Hg and absolute pressure is 2×10^3 mm of Hg.	M1.01	A
3	Name any one variable head flow meter.	M2.02	U
4	What is the use of pitot tube ?	M2.02	U
5	Name any one hydrostatic pressure type level gauge.	M3.02	U
6	Name any one type electrical type level gauge	M3.01	U
7	Name one non – contact type temperature measuring device.	M4.02	U
8	List any two types of resistance temperature detectors.	M4.02	R
9	Expand PTC and NTC.	M4.03	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	Express ‘1 atm.’ pressure in three different units.	M1.01	A
2.	Compare U tube manometer and well type manometer.	M1.02	U
3.	Draw the picture of rotameter and mark different parts.	M2.02	U
4.	Differentiate between laminar flow and turbulent flow	M2.01	U
5.	With figure describe sight glass type level indicator.	M3.01	U
6.	With figure ultrasonic type level gauge.	M3.02	U
7.	Differentiate between PTC and NTC.	M4.02	U
8.	Convert 50°C in to $^{\circ}\text{F}$ and $^{\circ}\text{K}$.	M4.01	A
9.	With figure describe vacuum measurement using Pirani gauge.	M1.05	U
10.	Explain Seebeck effect.	M3.01	U

Part – C

Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	With necessary figure, explain vacuum measurement using McLeod gauge.	M1.05	U
	OR		
IV	With necessary figure, describe the construction and working of C type Bourden tube pressure gauge.	M1.03	U
V	With necessary figure explain the flow measurement using electromagnetic flow meter.	M2.02	U
	OR		
VI	With necessary figure and equation explain the flow measurement using venturi meter.	M2.03	A
VII	With necessary figure and equation explain the flow measurement using orifice plate.	M2.03	A
	OR		
VIII	With necessary figure explain flow measurement using ultrasonic flow meter.	M2.02	U
IX	With necessary figure explain level measurement using displacer and torque tube.	M3.01	U
	OR		
X	With necessary figure and equation explain the capacitive type level gauge.	M3.01	U
XI	With necessary figure explain temperature measurement using optical pyrometers.	M4.02	U
	OR		
XII		M4.02	U

	With necessary figures explain construction and characteristics of RTD.		
XIII	State and explain the Peltier effect and Thomson effect.	M4.02	U
	OR		
XIV	With necessary figures explain construction of any two types of thermistors and thermistor characteristics.	M4.02	U

Question Wise Analysis

Course: **PROCESS VARIABLE MEASUREMENTS** – Question -2

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.01	U	1	2
I.2	M1.01	A	1	2
I.3	M2.02	U	1	2
I.4	M2.02	U	1	2
I.5	M3.02	U	1	2
I.6	M3.01	U	1	2
I.7	M4.02	U	1	2
I.8	M4.02	R	1	2
I.9	M4.03	R	1	2
II.1	M1.01	A	3	8
II.2	M1.02	U	3	8
II.3	M2.02	U	3	8
II.4	M2.01	U	3	8
II.5	M3.01	U	3	8
II.6	M3.02	U	3	8
II.7	M4.02	U	3	8
II.8	M4.01	A	3	8
II.9	M1.05	U	3	8
II.10	M3.01	U	3	8
III	M1.05	U	7	16
IV	M1.03	U	7	16
V	M2.02	U	7	16
VI	M2.03	A	7	16
VII	M2.03	A	7	16

VIII	M2.02	U	7	16
IX	M3.01	U	7	16
X	M3.01	U	7	16
XI	M4.02	U	7	16
XII	M4.02	U	7	16
XIII	M4.02	U	7	16
XIV	M4.02	U	7	16

Course : **PROCESS VARIABLE MEASUREMENTS –QP 2**

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Mark Distribution

Module	hr / Module	Marks/Module ($\frac{h_i}{\sum H_i} \times 123 (\pm 5\%)$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	15	30.5 ± 6	2	2	3	9	2	14	7	25
2	16	33 ± 6	2	2	2	6	4	28	8	36
3	12	24.5 ± 6	2	2	2	6	2	14	6	22
4	17	35 ± 6	3	3	3	9	4	28	10	40
Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	2	1.5
Understanding	100	81.5
Applying	21	17

Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100